

CIVIC-TECH HACKATHON

FRA MONITORING SYSTEM

From fragmented claims to faster action

Hackathon team :

- Gauri Singh
- Borancha Karthika
- Lasya Satti
- Urumu Chethan Sai



THE PROBLEM

The signal is buried in fragmented claim data

Claims, approvals, and land-use records live in separate workflows. The result is a shared problem with no shared operating view.

01

Separate workflows

State and district teams reconcile records by hand.

02

Late signals

Delayed cases and land mismatches surface late.

03

Case pile

Important exceptions become hard to see.

- WHY IT MATTERS

When monitoring is slow, rights decisions slow down too

Human cost

Applicants wait longer for clarity and resolution.

The delay is felt by people, not only systems.

Admin cost

Officers reconcile records instead of acting.

Time moves to comparison, not decision.

Environmental risk

Land-use discrepancies can remain invisible until late.

A visible exception can trigger earlier review.

Pilot metrics to validate: [CLAIMS IN PILOT] · [PENDING CASES] · [TARGET REVIEW-TIME REDUCTION]

THE OPERATING IDEA

One operating view for claims, land, and risk

Map the claim. Surface the exception. Give officials the next action.

FRA MONITOR · STATE VIEW

District overview



Map every claim

State · district · geography



Review alerts

Delay · mismatch · anomaly

FILTERS

STATE

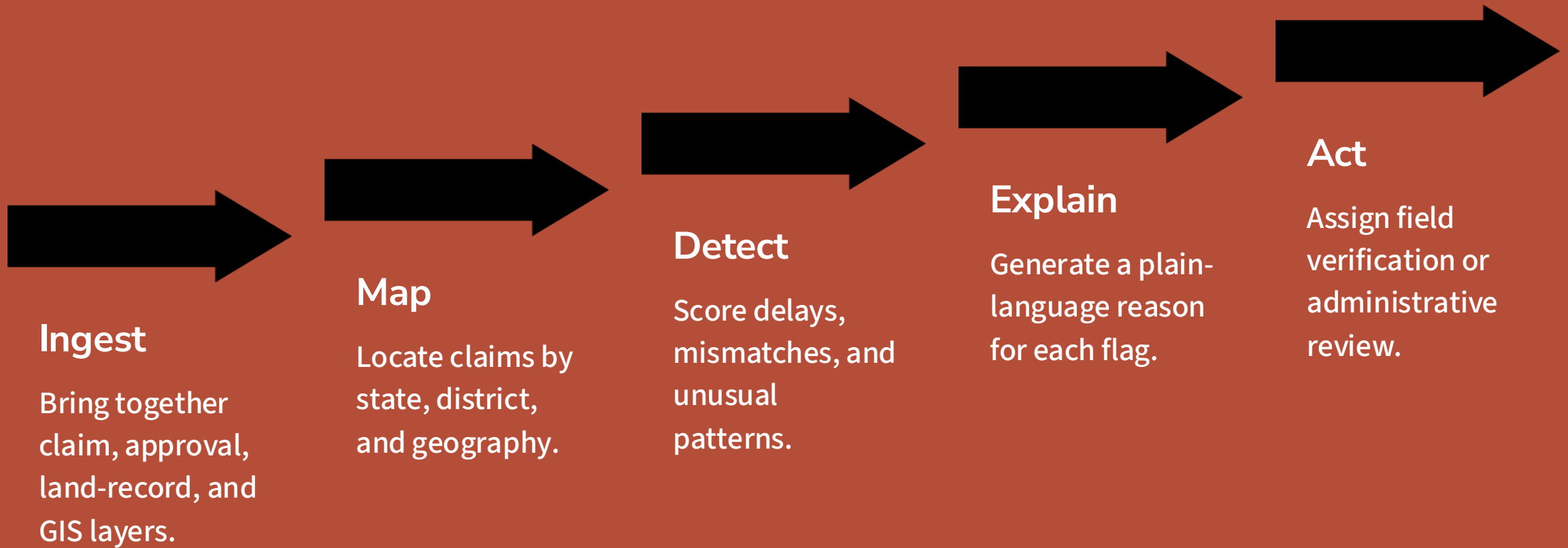
DISTRICT

NEXT ACTION

Claims, approvals, pending cases, and reasons in one view.

THE WORKFLOW

From raw records to a decision-ready alert



Four features turn monitoring into action



Interactive WebGIS map

Live claim and district progress view.



AI anomaly detection

Flags delayed claims and land-record mismatches.



Decision-support dashboard

Claims, approvals, pending cases, and alerts together.



AI alerts + explanations

Prioritizes cases and explains why they were flagged.

TECHNICAL FOUNDATION

A practical architecture built for government workflows

Modular layers keep the prototype understandable, deployable, and ready for verified public-sector data.

Frontend

React, HTML, CSS, JS, Leaflet for maps

Data Layer

PostgreSQL + PostGIS, GeoJSON, OpenStreetMap

API Layer

Python with FastAPI exposing REST endpoints

AI & Deployment

Scikit-learn anomaly detection, LLM explanations; GitHub, Vercel/Render, laptop



WHY THIS APPROACH

It combines geographic context with explainable prioritization

Dimension	Existing pattern	FRA Monitor
Context	Spreadsheets, maps, and manual review stay separate.	One map-led view with district filters.
Signal	Search and reconciliation surface problems.	Anomaly flags surface ranked cases.
Decision	Workflow reasoning stays implicit.	Each flag carries a plain-language reason.

INNOVATION

AI + WebGIS integration

SHIFT

Action-oriented insights

SCALE

Scalable decision support

DEMO SEQUENCE

Prototype flow: see the case, understand the flag, take the next step

01

SCREENSHOT
PLACEHOLDER

State / district overview

Map and progress cards.

02

SCREENSHOT
PLACEHOLDER

Claim detail

Mismatch or delay alert.

03

SCREENSHOT
PLACEHOLDER

Action panel

AI explanation and field-
verification action.

Demo sequence: filter district → open flagged claim → inspect map context → review explanation → assign action.

Start with one pilot, then scale the operating model

Pilot district

Connect available claim, land, and GIS layers. Validate alert logic with officers.

State rollout

Add district comparisons and role-based views. Establish repeatable data refreshes.

Cross-scheme expansion

Reuse the AI + GIS pattern for welfare and land-governance programs.

User target

Forest and tribal welfare officers · state/district administrators · field teams · rights committees

A sustainable path from hackathon prototype to public infrastructure

01 Deploy

Centralized dashboard for state and district FRA departments.

02 Adopt

Pilot district with a focused set of verified datasets.

03 Sustain

Government-focused SaaS/service model with annual maintenance, support, and customization.

04 Extend

Add GIS layers, data sources, and welfare-monitoring use cases over time.

A service that gets better with each verified layer.

Start narrow. Learn with officers. Expand the pattern, not the promise.

THE ASK

Make every FRA claim visible, explainable, and actionable

FRA MONITORING SYSTEM turns fragmented records into a shared operating picture for faster, more transparent decisions.

01 Build with us

Pilot the workflow with one district or state.

02 Measure

Track review time, flagged-case resolution, and data-quality improvements.

03 Scale

Take the AI + WebGIS pattern to the next public-service challenge.